

6. Applications of Integrals

6.5. Average Value of a Function

6.5 Learning Objectives:

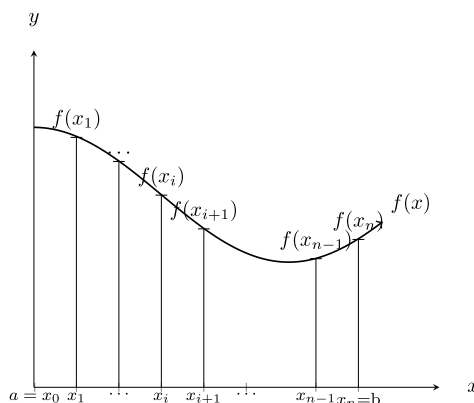
- I can find the average value of a function on some interval.
- I can know and understand how to apply the Mean Value Theorem.

Definition 6.5.1

Recall the average of n values, namely $a_1, a_2, \dots, a_n$, is

$$\text{average} = \frac{a_1 + a_2 + a_3 + \dots + a_n}{n} = \sum_{i=1}^n \frac{1}{n} a_i$$

Suppose we wanted to find the average of all values that a function $f(x)$ achieves on an interval $[a, b]$. We can think about sampling the function at n equally spaced points and dividing by the number of points. As we look at sampling more points, we get a more precise approximation for the average of f .



The approximate function average is

$$\begin{aligned} f_{\text{ave}} &\approx \frac{f(x_1) + f(x_2) + \dots + f(x_n)}{n} \\ &= \frac{1}{n} \sum_{i=1}^n f(x_i) \\ &= \frac{\Delta x}{b-a} \sum_{i=1}^n f(x_i) \\ &= \frac{1}{b-a} \sum_{i=1}^n f(x_i) \Delta x \end{aligned}$$

To make our approximation more accurate, we will sample the function at even more points, eventually taking the limit gives us the integral.

$$\begin{aligned}
 f_{\text{ave}} &= \frac{1}{b-a} \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x \\
 &= \frac{1}{b-a} \int_a^b f(x) \, dx.
 \end{aligned}$$

Definition 6.5.2

The average value of f on $[a, b]$ is given by

$$f_{\text{ave}} = \frac{1}{b-a} \int_a^b f(x) \, dx.$$

Example 6.5.1

Find the average of $f(x) = 1 + x^2$ on $[-1, 2]$.

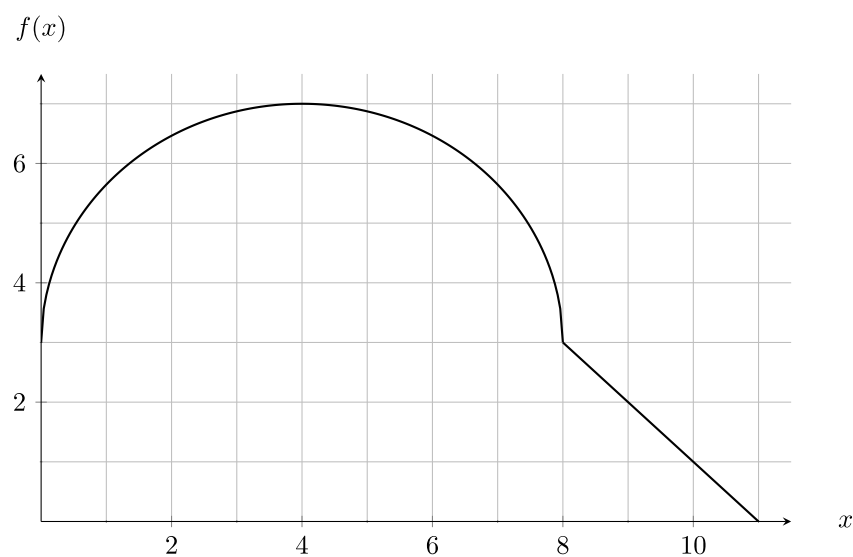
Solution

We use the formula

$$\begin{aligned}
 f_{\text{ave}} &= \frac{1}{2 - (-1)} \int_{-1}^2 1 + x^2 \, dx \\
 &= \frac{1}{3} \left[x + \frac{x^3}{3} \right]_{-1}^2 \\
 &= \frac{1}{3} \left[2 + \frac{8}{3} - \left(-1 - \frac{1}{3} \right) \right] \\
 &= 2
 \end{aligned}$$

Example 6.5.2

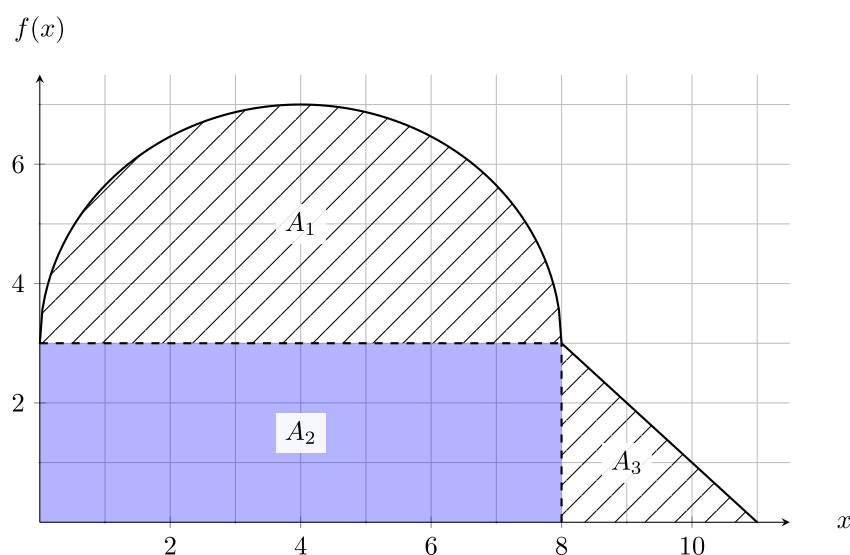
Find the average value of f , given in the graph below, on $[0, 11]$.

**Solution**

We use the formula for average of a function

$$f_{\text{ave}} = \frac{1}{11-0} \int_0^{11} f(x) \, dx$$

In this case, to get the integral, we remember that the integral of a curve is the same as the area under that curve. We don't need the equation for f since we can use geometry to find the area.



Now, we add the areas of the different parts up: $A_1 = \frac{1}{2}\pi r^2 = \frac{1}{2}\pi 4^2 = 8\pi$ And $A_2 = wh = (8)(3) = 24$. And $A_3 = \frac{1}{2}bh = \frac{1}{2}(3)(3) = \frac{9}{2}$.

Then, the average is

$$\begin{aligned}
 f_{\text{ave}} &= \frac{1}{11} \int_0^{11} f(x) \, dx \\
 &= \frac{1}{11} (A_1 + A_2 + A_3) \\
 &= \frac{1}{11} \left(8\pi + 24 + \frac{9}{2} \right).
 \end{aligned}$$

Theorem 6.5.1 Mean Value Theorem for Integrals

If f is continuous on $[a, b]$, then there is some c in (a, b) where

$$f(c) = f_{\text{ave}} = \frac{1}{b-a} \int_a^b f(x) \, dx$$

Example 6.5.3

Does the function $f(x) = (x+1)^3$ achieve its average value on $[a, b]$ for some value in (a, b) ? If so where does this happen?

Solution

Yes, because f is continuous (since it is a polynomial), so by Mean Value Theorem (MVT) the function achieves its average value for some c in (a, b) .

As for where this happens, the MVT does not say. We can find it though,

$$\begin{aligned}
 f_{\text{ave}} &= \frac{1}{2-0} \int_0^2 (x+1)^3 \, dx \\
 &= \frac{1}{2} \int_1^3 u^3 \, du \\
 &= \frac{1}{2} \frac{u^4}{4} \Big|_1^3 \\
 &= \frac{1}{2} \left(\frac{3^4}{4} - \frac{1}{4} \right) \\
 &= 10
 \end{aligned}$$

Where the u -sub we used was $u = x + 1$ so $du = dx$ and $u(0) = 1$ and $u(2) = 3$.

Since the average value is now known (10), we can find which c value has $f(c) = 10$.

$$\begin{aligned}
 10 &= (c+1)^3 \\
 \sqrt[3]{10} &= c+1 \\
 \sqrt[3]{10} - 1 &= c
 \end{aligned}$$

which is in $(0, 2)$.

6.5 Section Summary:

- We found the average value of a function on some interval.
- We used mean value theorem to find where a function achieves its average value.